

KONGU ENGINEERING COLLEGE

SMART IRRIGATION AND WATER MANAGEMENT SYSTEM MECHATRONICS AND IOT

EXCERSICE - 04

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1. Project Overview

The Smart Irrigation and Water Management System is an automated agricultural system developed to provide efficient irrigation based on the actual condition of the soil and surrounding environment. Traditional irrigation methods often depend on manual operation, which can result in over-irrigation, water wastage, or insufficient watering. This project aims to overcome these problems by using sensors, a microcontroller, and an automatic water pumping mechanism.

The system uses an ESP8266 NodeMCU as the main control unit. A soil moisture sensor continuously measures the moisture content of the soil. When the soil becomes dry and the moisture level falls below the predefined threshold, the NodeMCU processes the sensor information and activates the water pump through a relay module. The pump supplies water to the plants until the required moisture condition is achieved. The pump is then automatically switched OFF to prevent unnecessary water consumption.

The IoT capability of the ESP8266 allows the collected sensor information to be transmitted through a Wi-Fi network to a cloud monitoring platform. The user can monitor soil moisture, temperature, humidity, water level, and pump status remotely. The recorded data can also be displayed using graphs, making it easier to observe changes in environmental conditions and irrigation activity over time.

Overall, the Smart Irrigation and Water Management System provides a simple, economical, and efficient solution for automatic irrigation. It combines sensor-based monitoring, automatic pump control, and IoT-based data monitoring to support better water management and modern agricultural practices.

MAIN FEATURES

- Automatic irrigation based on soil moisture level.
- Continuous monitoring of soil moisture, temperature, and humidity.
- Water-level monitoring to check the availability of water in the tank.
- Automatic ON/OFF control of the water pump using a relay.
- IoT-based monitoring using ESP8266 and Wi-Fi.
- LED or buzzer indication for low moisture and low water-level conditions.
- Reduces water wastage by supplying water only when required.
- Reduces manual effort and provides continuous monitoring.
- Helps maintain suitable moisture conditions for healthy plant growth.
- Suitable for agricultural fields, gardens, nurseries, and greenhouses.

2. PROBLEM STATEMENT

Traditional irrigation systems commonly depend on manual operation or fixed irrigation schedules. These methods do not consider the actual moisture condition of the soil.

This can result in:

- Excessive water usage.
- Under-irrigation of plants.
- Over-irrigation and waterlogging.
- Unnecessary operation of water pumps.
- Increased electricity consumption.
- Requirement for continuous human monitoring.

Therefore, an automated system is required to monitor soil conditions and provide water only when it is necessary

3. PROBLEM SOLUTION

The proposed Smart Irrigation and Water Management System provides an automatic method for controlling irrigation based on real-time soil and environmental conditions. An ESP8266 NodeMCU is used as the main controller to collect data from the soil moisture sensor, DHT11 temperature and humidity sensor, and water-level sensor.

When the soil moisture level falls below the predefined threshold, the NodeMCU checks the available water level and automatically turns ON the water pump through a relay. Water is supplied to the plants until the required soil moisture level is reached. The pump is then switched OFF automatically.

The ESP8266 can also connect to a Wi-Fi network and send the collected data to an IoT cloud platform. This allows users to monitor soil moisture, temperature, humidity, water level, and pump status remotely. The proposed solution reduces water wastage, minimizes manual effort, and provides efficient irrigation management for agricultural applications.

When the soil becomes dry:

Soil Moisture Sensor → ESP8266 → Relay ON → Water Pump ON

When sufficient moisture is reached:

Soil Moisture Sensor → ESP8266 → Relay OFF → Water Pump OFF

This automatic control helps provide water according to the requirement of the plants

4. SYSTEM ARCHITECTURE

The system architecture consists of sensing, control, irrigation, and IoT monitoring sections. The ESP8266 NodeMCU acts as the main controller and receives information from all the connected sensors.

The soil moisture sensor measures the moisture level of the soil, while the DHT11 sensor measures temperature and humidity. The water-level sensor monitors the amount of water available in the storage tank. The sensor data is sent to the NodeMCU for processing and decision-making.

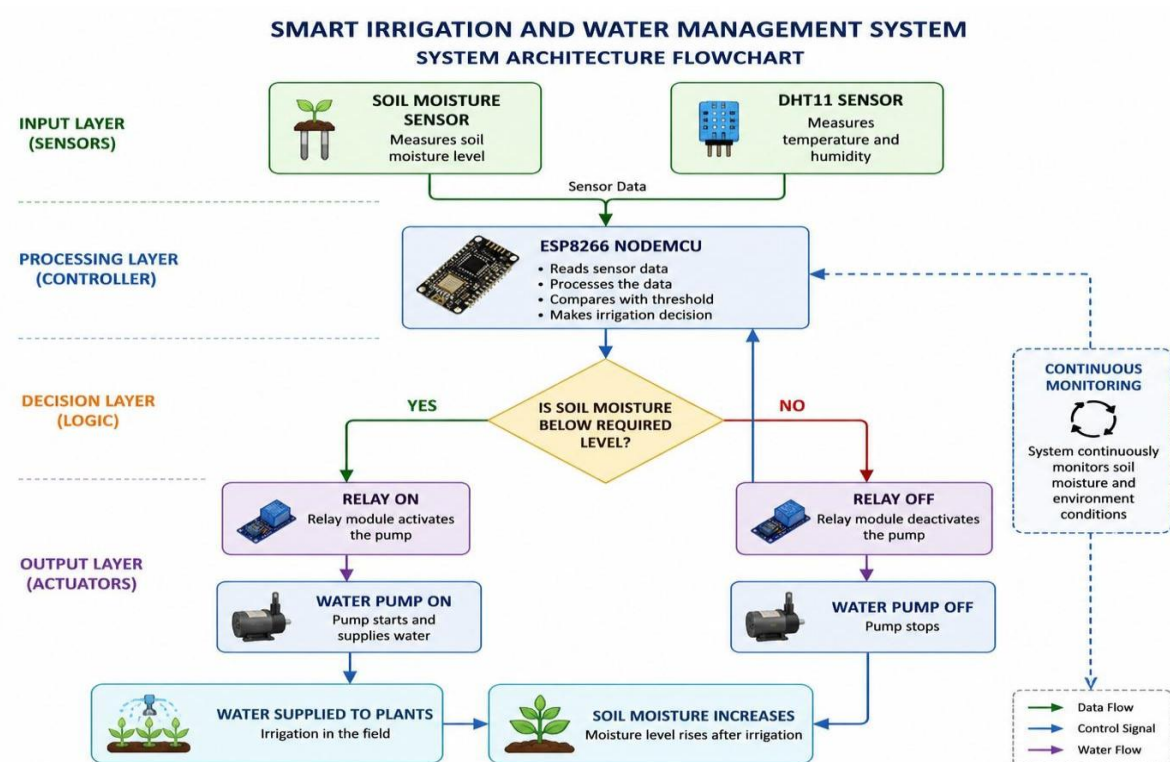
Based on the sensor readings, the NodeMCU controls the water pump through a relay module. When the soil is dry and sufficient water is available, the pump is switched ON. When the required moisture level is reached, the pump is switched OFF automatically.

For IoT monitoring, the NodeMCU uses Wi-Fi to transmit the collected data to a cloud platform. The user can monitor the environmental conditions and pump status remotely.

System flow:

Sensors → ESP8266 NodeMCU → Relay Module → Water Pump → Irrigation

ESP8266 NodeMCU → Wi-Fi → Cloud Platform → Remote Monitoring



5. CIRCUIT TABLE

Component	Pin	ESP8266 NodeMCU
Soil Moisture Sensor	VCC	3V3
Soil Moisture Sensor	GND	GND
Soil Moisture Sensor	AO	A0
DHT11	VCC	3V3
DHT11	GND	GND
DHT11	DATA	D2
Relay Module	VCC	VIN
Relay Module	GND	GND
Relay Module	IN	D3
Relay Module	COM	Pump power supply +
Relay Module	NO	Pump +
Water Pump	–	Pump power supply –

The circuit is built around the ESP8266 NodeMCU, which acts as the main controller. The soil moisture sensor is connected to the NodeMCU to measure the moisture level of the soil. The DHT11 sensor is connected to a digital pin to measure temperature and humidity. A water-level sensor is used to monitor the water available in the storage tank. The relay module is connected to the NodeMCU and is used to control the water pump. When the soil becomes dry, the NodeMCU activates the relay and switches ON the pump. When the required moisture level is reached, the relay switches OFF the pump. The sensors and controller share a common ground, while the water pump is operated using a suitable external power source through the relay. The ESP8266 can also connect to Wi-Fi for transmitting sensor data to an IoT cloud platform for remote monitoring.

Power Supply

The ESP8266 can be powered through USB. The relay is powered from the appropriate 5V supply through VIN when the board is USB powered.

The water pump should use a suitable power supply according to its rated voltage and current. For a 5V pump, a regulated 5V supply or suitable boost converter should be used.

6. CIRCUIT DIAGRAM

The circuit consists of five major sections:

6.1 ESP8266 NodeMCU

The ESP8266 is the main controller. It receives sensor data and controls the relay according to the soil moisture condition.

6.2 Soil Moisture Sensor

The soil moisture sensor is connected to the analog input A0.

It provides an analog value representing the moisture condition of the soil. Soil

Sensor AO → ESP8266 A0

6.3 DHT11 Sensor

The DHT11 measures:

- Temperature.
- Relative humidity.

DHT11 DATA → ESP8266 D2

6.4 Relay Module

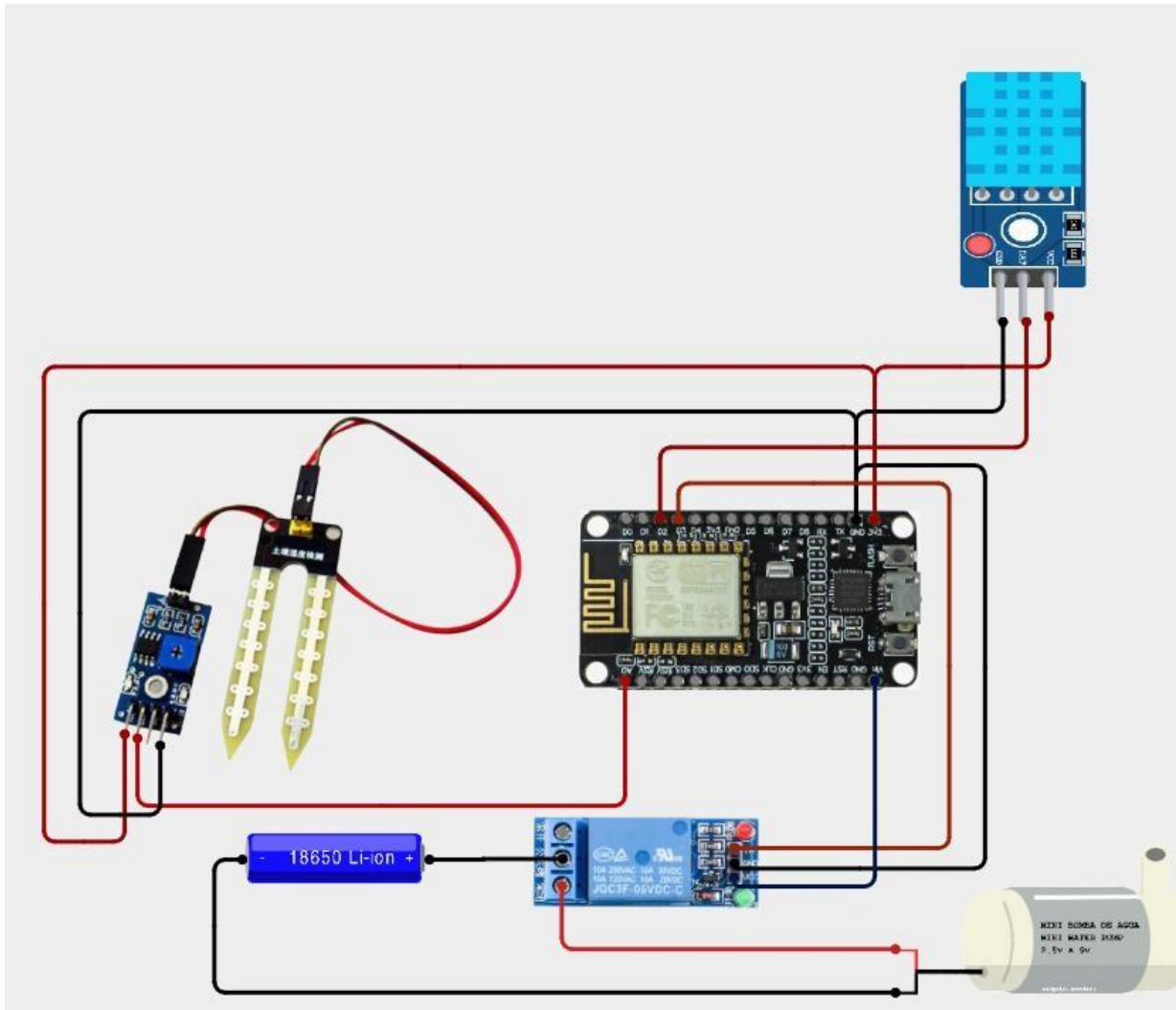
The relay provides electrical switching between the low-power controller and the water pump.

Relay IN → ESP8266 D3

6.5 Water Pump

The water pump supplies water to the plants. It is controlled through the relay and should not be connected directly to an ESP8266 GPIO pin.

CIRCUIT DSIGN

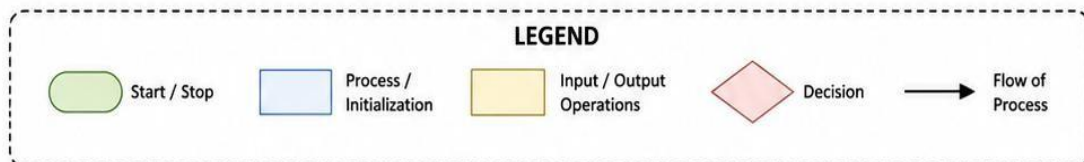
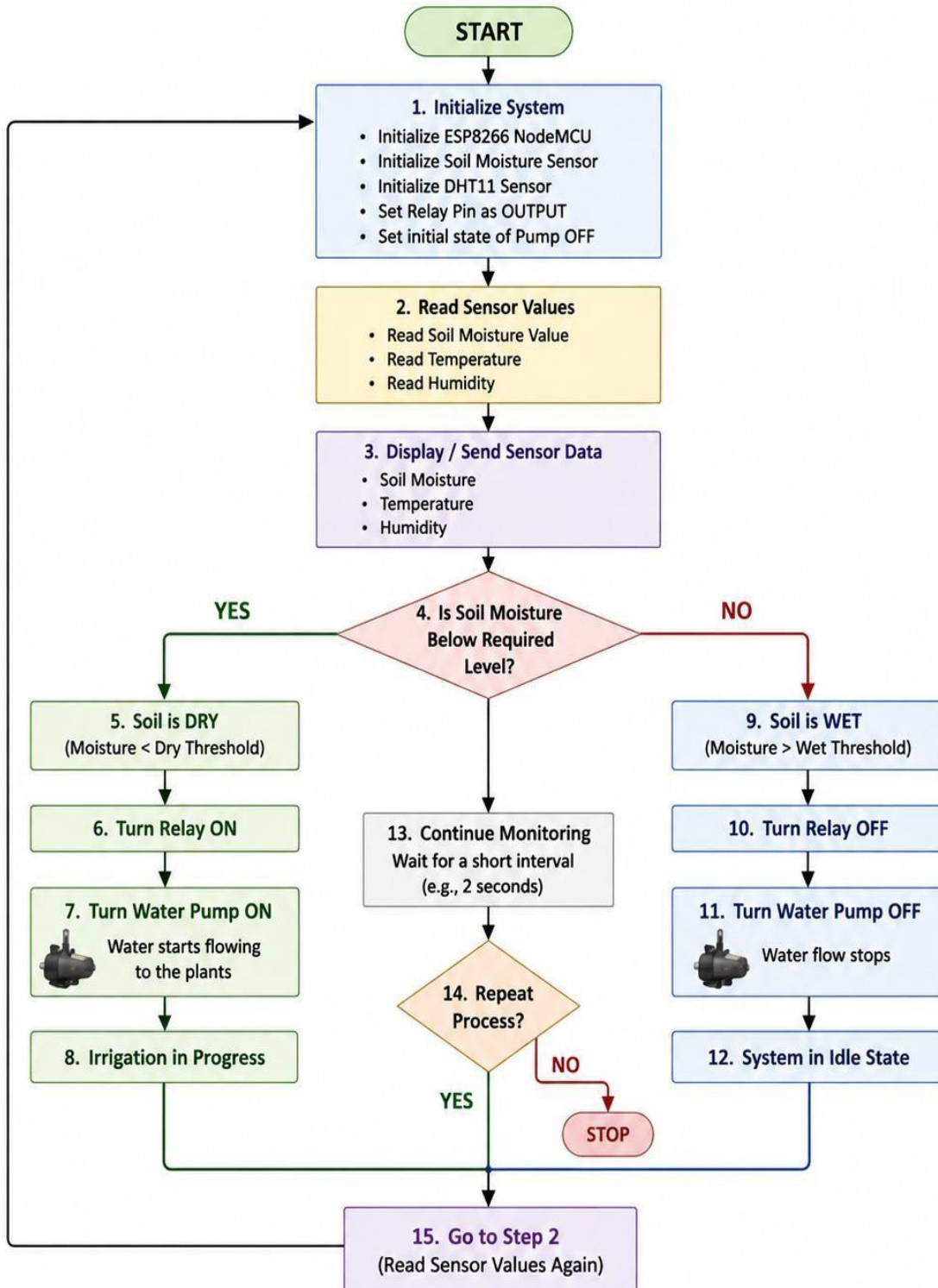


7. ALGORITHM

The system starts by initializing the ESP8266 NodeMCU, soil moisture sensor, DHT11 sensor, water-level sensor, relay module, and water pump. The system then continuously reads the soil moisture, temperature, humidity, and water-level values. The NodeMCU compares the soil moisture value with the predefined threshold. If the soil is dry and sufficient water is available in the tank, the relay is activated and the water pump is switched ON. The pump continues to supply water until the soil reaches the required moisture level. Once the required moisture level is reached, the pump is switched OFF automatically. If the water level is too low, the pump remains OFF to prevent damage. The sensor readings and pump status can also be sent through Wi-Fi to an IoT cloud platform for remote monitoring. This process is repeated continuously to provide automatic and efficient irrigation.

SMART IRRIGATION AND WATER MANAGEMENT SYSTEM

ALGORITHM FLOWCHART



8. CODING

The following Arduino program is designed for the ESP8266 NodeMCU used in the project.

```
#include <DHT.h>

#define SOIL_PIN A0
#define DHT_PIN D2
#define RELAY_PIN D3

#define DHT_TYPE DHT11

DHT dht(DHT_PIN, DHT_TYPE);

int dryThreshold = 700;
int wetThreshold = 500;

void setup() {

    Serial.begin(115200);

    pinMode(SOIL_PIN, INPUT);
    pinMode(RELAY_PIN, OUTPUT);

    // Pump OFF initially
    digitalWrite(RELAY_PIN, HIGH);

    dht.begin();

    Serial.println("Smart Irrigation System Started");
}
```

```
void loop() {

    int soilValue = analogRead(SOIL_PIN);

    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    Serial.println("-----");

    Serial.print("Soil Moisture: ");
    Serial.println(soilValue);

    if (isnan(temperature) || isnan(humidity)) {

        Serial.println("DHT11 reading failed");

    } else {

        Serial.print("Temperature: ");
        Serial.print(temperature);
        Serial.println(" C");

        Serial.print("Humidity: ");
        Serial.print(humidity);
        Serial.println(" %");
    }

    if (soilValue >= dryThreshold) {
```

```
digitalWrite(RELAY_PIN, LOW);

Serial.println("Soil Status: DRY");
Serial.println("Pump Status: ON");

}

else if (soilValue <= wetThreshold) {

digitalWrite(RELAY_PIN, HIGH);

Serial.println("Soil Status: WET");
Serial.println("Pump Status: OFF");

}

else {

Serial.println("Soil Status: MODERATE");
Serial.println("Pump Status: Previous State");

}

delay(2000);
}
```

CODING EXPLANATION

The Arduino program is developed for the ESP8266 NodeMCU to control the irrigation system automatically based on soil moisture. It also monitors temperature and humidity using the DHT11 sensor.

At the beginning, the required DHT library is included and the pins for the soil moisture sensor, DHT11 sensor, and relay are defined. The soil moisture sensor is connected to A0, the DHT11 sensor to D2, and the relay to D3. Two threshold values are defined: 700 for dry soil and 500 for wet soil.

When the NodeMCU starts, the setup function initializes serial communication, sensors, and the relay. The relay is initially kept OFF so that the water pump does not operate immediately after power-up.

During the main loop, the NodeMCU reads the soil moisture value from the soil moisture sensor. It also reads temperature and humidity from the DHT11 sensor. The readings are displayed on the Serial Monitor for monitoring.

If the soil moisture value is 700 or above, the soil is considered dry. The NodeMCU activates the relay and turns ON the water pump. If the value is 500 or below, the soil is considered wet, so the relay is deactivated and the pump is switched OFF. When the value is between 500 and 700, the soil is considered moderately moist and the previous pump condition is maintained.

The program waits for 2 seconds before taking the next readings. This process is continuously repeated, allowing the system to automatically supply water when required and stop irrigation when sufficient moisture is available. Thus, the program helps reduce water wastage and manual effort while maintaining suitable soil moisture for plants.

9. SYSTEM WORKING

The Smart Irrigation and Water Management System works by continuously monitoring the soil and environmental conditions. The ESP8266 NodeMCU acts as the main controller and receives data from the soil moisture sensor and DHT11 sensor. The soil moisture sensor determines whether the soil is dry, moderate, or wet, while the DHT11 measures temperature and humidity.

When the soil moisture level falls below the predefined dry threshold, the NodeMCU activates the relay and switches ON the water pump. The pump supplies water to the plants until the soil reaches the required moisture level. When the soil becomes sufficiently wet, the NodeMCU switches OFF the relay and stops the pump.

The temperature and humidity readings are displayed on the Serial Monitor for monitoring. The system repeats the sensing and control process at regular intervals. In an IoT-enabled version, the sensor data can also be transmitted through Wi-Fi to a cloud platform for remote monitoring. This automatic operation helps reduce water wastage, minimize manual effort, and improve irrigation efficiency.

Dry Soil Condition

When the soil moisture value indicates that the soil is dry:

Soil Dry



ESP8266 detects dry condition



Relay ON



Water Pump ON



Water supplied to plants

Wet Soil Condition

When the soil reaches the required moisture level:

Soil Wet



ESP8266 detects sufficient moisture



Relay OFF



Water Pump OFF

At the same time, the DHT11 continuously measures temperature and humidity.

This process repeats automatically, allowing the system to maintain suitable soil moisture.

10. BILL OF MATERIALS

S.No.	Component	Quantity
1	ESP8266 NodeMCU	1
2	Soil Moisture Sensor	1
3	1-Channel Relay Module	1
4	Mini Water Pump	1
5	DHT11 Sensor	1
6	Breadboard	1
7	Jumper Wires	1 set
8	Suitable Pump Power Supply	1

SOFTWARE REQUIRED

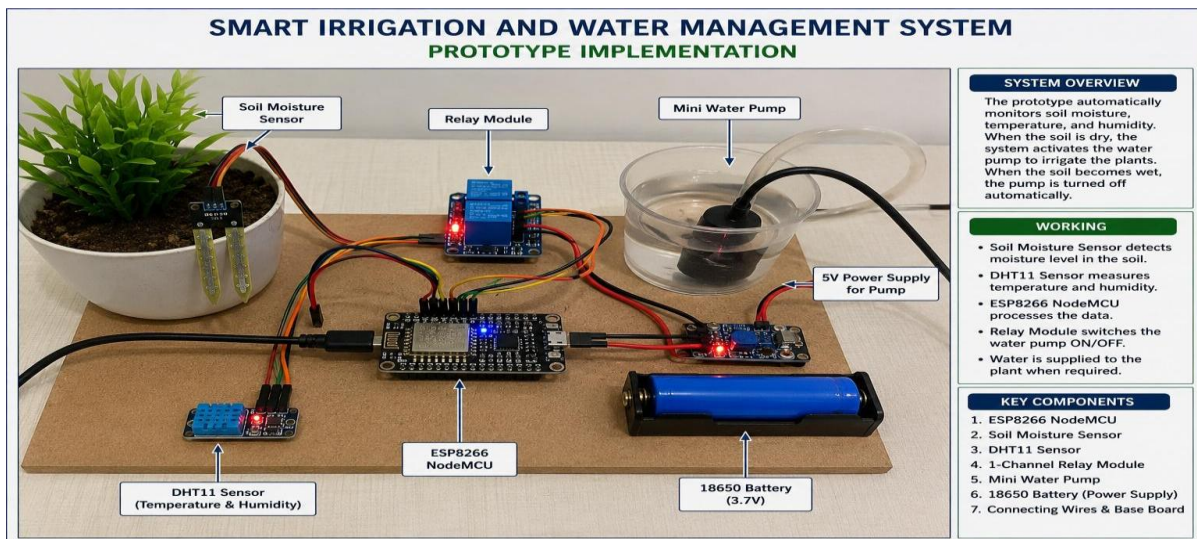
- Arduino IDE
- ESP8266 Board Package
- DHT Sensor Library
- Cirkuit Designer for circuit design/simulation

11. PROTOTYPE IMPLEMENTATION

The prototype of the Smart Irrigation and Water Management System is implemented using an ESP8266 NodeMCU, soil moisture sensor, DHT11 sensor, relay module, DC water pump, water container, and connecting wires. The NodeMCU is used as the main controller to collect sensor readings and control the irrigation process.

The soil moisture sensor is placed in the soil to measure the moisture level. The DHT11 sensor measures the surrounding temperature and humidity. The relay module is connected to the NodeMCU to control the water pump. When the soil becomes dry, the NodeMCU activates the relay and the pump supplies water to the soil. Once sufficient moisture is detected, the pump is automatically switched OFF.

The prototype is tested under different soil moisture conditions to verify the automatic operation of the pump. The sensor values and pump status are displayed on the Serial Monitor. The prototype demonstrates that irrigation can be controlled automatically based on real-time soil conditions, helping to reduce water wastage and manual intervention.



12. RESULT & PERFORMANCE AYALYSIS

Expected Results

The system successfully demonstrates automatic irrigation based on soil moisture.

Condition	Sensor Condition	Relay	Pump
Dry soil	Moisture below required level	ON	ON
Moderate soil	Intermediate value	Maintains previous state	Maintains previous state
Wet soil	Moisture sufficient	OFF	OFF

PERFORMANCE

The system provides the following benefits:

- Automatic irrigation.
- Reduced unnecessary water usage.
- Reduced manual intervention.
- Prevention of over-irrigation.
- Continuous environmental monitoring.
- Automatic pump control.
- Potential for remote IoT monitoring

The Smart Irrigation and Water Management System was successfully implemented and tested using the ESP8266 NodeMCU, soil moisture sensor, DHT11 sensor, relay module, and water pump. The system was able to detect different soil moisture conditions and automatically control the irrigation pump.

When the soil moisture value reached the dry threshold, the pump was switched ON and water was supplied to the soil. When sufficient moisture was detected, the pump was automatically switched OFF. The DHT11 sensor successfully measured temperature and humidity, and the readings were displayed on the Serial Monitor.

The prototype provided reliable automatic irrigation control and reduced unnecessary water usage. The system also reduced manual effort by continuously monitoring the soil and controlling the pump automatically. Overall, the prototype demonstrated effective and efficient irrigation management for small-scale agricultural applications.

